GPU Acceleration of a Diabetic Retinopathy Classification Pipeline

Final technical report — GPU Programming and Performance Optimization, 2026

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Rule followed throughout: every number in this report is read from a file in `results/`; the file is named next to the number and the command that produced it is in section 11. Week-level details are in `report/week1_report.md` (baseline, profiling), `week2_report.md` (cache, GPU image processing) and `week3_report.md` (GPU pipeline, model redesign). This report adds the Week 4 work (GPU profiling, ablation, test-set evaluation, same-machine end-to-end benchmark) and puts the four weeks together.

1. Abstract and objectives

We profiled, redesigned and GPU-accelerated a CPU reference pipeline that classifies 2,999 retinal fundus images (FDR26) into five diabetic-retinopathy grades. The **principal bottleneck of the baseline was image loading** (JPEG decode + resize, **65.4 %** of a profiled CPU epoch, `results/profile_stages.json`), which bounds any speedup that leaves loading untouched at 2.89× (Amdahl). We removed it with a decoded-image cache (CPU, 1,807× on the stage), moved preprocessing to batched GPU kernels, kept the whole dataset resident in GPU memory, and trained with mixed precision and `channels_last`. We then redesigned the model (ResNet-18 pretrained, cosine schedule, 320 px input, batch 64).

Main measured results:

- **End-to-end, same machine (Colab, Tesla T4 + 1-core CPU): 60.7× faster** for the full 5-epoch training (1,427.5 s → 23.5 s) and **65.4× per steady epoch** (285.4 s → 4.36 s), `results/week4/end_to_end.csv`.
- **Test set (used once, 3 seeds): macro-F1 0.313 ± 0.010 → 0.590 ± 0.021**, accuracy 0.357 → 0.598, `results/week4/test_summary.csv`. Most remaining errors are between adjacent grades.
- **Final GPU profile: the GPU is idle 0.4 % of the time** and 0.6 % of a step is data handling: the pipeline is now limited by the T4's convolution, batch-norm and pooling kernels, not by the input path (`results/week4/profile_optimized/summary.json`).
- **Ablation:** mixed precision is the most valuable single optimization of the final design (4.54×), then `channels_last` (1.31×), GPU-resident data (1.23×) and precompute (1.17×); cuDNN autotuning and the fused optimizer give nothing measurable (1.00×, 1.01×) — `results/week4/ablation.csv`.

The most important conclusion: **each optimization paid only while it targeted the current bottleneck**. The bottleneck moved from JPEG decoding (CPU) to CPU preprocessing, then to the CPU CNN, then to GPU preprocessing, and finally to GPU convolution compute, where it is now.

2. Problem and pipeline description

Task. 5-class DR grading (ALFEDIAM grades 0–4). FDR26 (Zenodo 20605487, CC BY 4.0): 3,000 images listed, 2,999 present (`2T2SALA702` is missing from the archive), source resolution 4.9–10.5 Mpx JPEG. **Splits** (`splits/splits.csv`, versioned, identical for every run): patient-level StratifiedGroupKFold, seed 42, 70/15/15 → **2,099 / 450 / 450** images; no patient appears in two splits (checked in Week 1). Classes are balanced inside each split.

CPU reference pipeline (modules A–E of the brief, diagram in `docs/pipeline.md`):

Module	Function	Work per image
A Loading	`DRDataset.load`	PIL decode of a ~2,576×1,958 JPEG, bilinear resize to 224×224
B Preprocessing	`preprocess_image`	CLAHE on L of LAB (clip 2, 8×8 tiles), circular crop, ImageNet normalization
C Filters	`filters_cpu.py`	Gaussian, median, green channel, contrast stretch (benchmarked; not on the training path)
D Augmentation	`augment_image`	flip + rotation ≤ 10° (off in the baseline)
E Training	`ReferenceCNN`, `run_epoch`	3 conv blocks + MLP, 0.62 M parameters, 1.01 GFLOPs/image, AdamW 1e-3, 5 epochs, batch 32

Model selection always uses the **validation** accuracy (best epoch checkpoint); the test set is used only by `scripts/week4_test_eval.py` (section 7.3).

3. Experimental environment (Table 1)

Two machines were used. **Every speedup in this report compares two runs on the same machine**; when a table mixes machines, the machine is written next to the number.

Item	Machine L (Weeks 1-2)	Machine C (Weeks 3-4)

CPU	AMD Ryzen 7 8845HS, 8 cores / 16 threads	Colab VM, **1 physical core / 2 threads**
RAM	16 GB	14 GB
GPU	RTX 4070 Laptop, 8 GB, CC 8.9, 36 SMs	**Tesla T4, 15.6 GB, CC 7.5, 40 SMs**, driver 580.82
OS	Windows 11 Home	Linux 6.6 (Colab)
Python / PyTorch / CUDA / cuDNN	3.14.6 / 2.14.1+cu130 / 13.0 / 9.24	3.13.16 / 2.11.0+cu130 / 13.0 / 9.27
OpenCV / NumPy / scikit-learn	5.0.0 / 2.5.2 / 1.9.1	5.0.0 / 2.1.3 / 1.6.1
Source	`environment/hardware_software.md`	`results/week4/environment.json`

Configuration	Baseline (`configs/baseline.yaml`)	Final (`configs/optimized.yaml`)
Model	reference CNN (0.62 M params)	ResNet-18, ImageNet weights (11.2 M params)
Input	224×224, CPU CLAHE/crop/normalize	320×320, same preprocessing on the GPU, computed once
Batch / epochs / optimizer	32 / 5 / AdamW 1e-3, no schedule	64 / 5 / fused AdamW 1e-3, cosine schedule
Precision / layout	fp32 / NCHW	AMP (fp16 + GradScaler) / `channels_last`
Data path	DataLoader, 2 workers, decode every epoch	whole split resident in GPU memory
Splits	2,099 / 450 / 450, patient-level, identical	identical

4. Baseline profiling

Methodology. `cProfile` on one CPU epoch with `num_workers=0` so that decoding is visible in the main process (`benchmarks/profile_baseline.py`), manual stage timers on 64 images (`benchmarks/benchmark_stages.py`), `line_profiler` and the PyTorch Profiler on the CPU training step, and 1 s CPU/RAM sampling of the real 2-worker run (`benchmarks/resource_monitor.py`). Machine L.

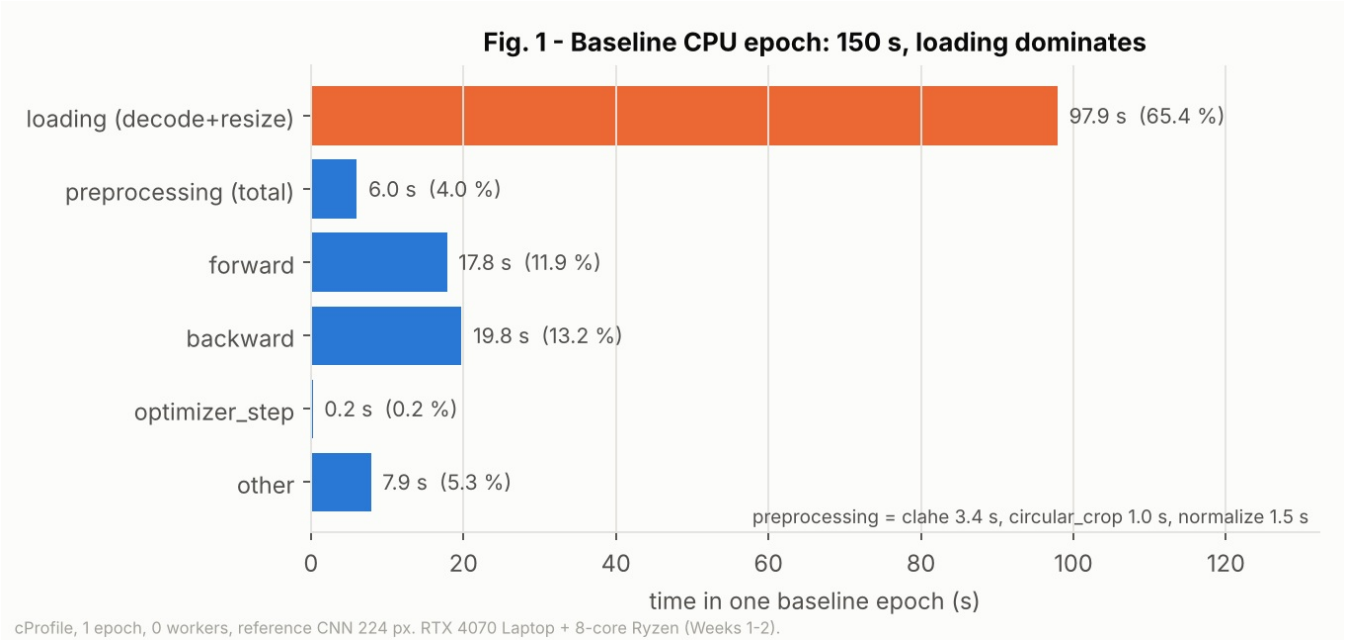


Figure 1 — Baseline runtime breakdown (one profiled CPU epoch, 149.6 s).

Rank	Stage	Time (s)	Share	Amdahl bound 1/(1−p)	Evidence
1	Loading (decode + resize)	97.9	**65.4 %**	2.89×	`results/profile_stages.json`
2	Backward	19.8	13.2 %	—	same
3	Forward	17.8	11.9 %	(fwd+bwd+opt) 1.34×	same
4	Other (Python, collate, conversions)	7.9	5.3 %	—	same
5	Preprocessing (CLAHE 3.4 s, crop 1.0 s, normalize 1.5 s)	6.0	4.0 %	1.04×	same
6	Optimizer step	0.2	0.2 %	—	same

Findings that decided the plan (details in `report/week1_report.md`): JPEG decode (39.4 s) and resize (42.0 s) of a 5-Mpx frame to 0.05 Mpx dominate; the disk is *not* the problem (file read 1.1 % of loading with a warm page cache); the real 2-worker baseline is loader-bound (loading alone 56.4 s per training epoch vs 28.7 s of compute). Baseline record (machine L): **85.3 s per epoch, 428.3 s for 5 epochs, validation accuracy 0.391, macro-F1 0.350** (`results/baseline/history.json`, `metrics_val.json`).

5. Redesign and optimization decisions

Each change is linked to the measurement that motivated it, applied alone first (`configs/exp_*.yaml`), validated, and then re-profiled.

#	Change	Motivation (measured)	Expected	Measured	Trade-off
1	**Decoded-image cache** (decode + resize once into a memory-mapped uint8 `.npy`)	loading = 65.4 % (sec. 4)	≤ 2.89× end-to-end	loading 1,807× (64 img, `timing.csv`); 5-epoch CPU run 428 → 281 s (1.53×)	+0.45 GB disk at 224 px; one-time build 21 s; cache keyed by (image list, size)
2	**Batched GPU preprocessing** (OpenCV-exact LAB CLAHE, one `scatter_add_` for all tiles)	once loading is gone, CPU CLAHE is the next stage	10–20× on the stage	17.3× (64 img), 15.4× (128 img); **bit-exact**	custom code, needs CUDA
3	**uint8 host→device copies** instead of float32	4× fewer bytes per batch	≤ 4× on the copy	3.9× (`results/week2/h2d.csv`)	—
4	**Training on the GPU**	CPU CNN = 46 img/s after (1)	> 10×	device effect alone 25.6× per epoch (machine C)	—
5	**GPU-resident data + precompute**	after (4) the loader (1 CPU core on Colab) feeds the GPU	remove all per-epoch PCIe traffic	ablation: 1.23× and 1.17× (sec. 7.4)	+1.5 GB GPU memory (fp16 tensors of all splits)
6	**AMP + `channels_last` + cuDNN benchmark + fused AdamW**	once compute-bound, use Tensor Cores	2×	2.3× on the reference CNN (Week 3); AMP alone 4.54× on ResNet-18 at 320 px	numerical: fp16 activations
7	**Model redesign**: ResNet-18 pretrained, cosine LR, 320 px, batch 64	reference CNN plateaus at val F1 ≈ 0.33	+ quality	val F1 0.33 → 0.58; test 0.31 → 0.59	3.6× FLOPs/image at 224 px, 2× pixels at 320 px
8	Prefetcher (side-stream copies)	overlap copies	small	no gain (3.92 → 3.97 s)	kept out of the final design
9	`torch.compile` / CUDA graphs	launch overhead	gain on small models	1.64× on the reference CNN but +21 s cold epoch; nothing on ResNet-18	kept off

Redesign challenge (brief, section 6), studied one change at a time with 3 seeds (`report/week3_report.md`, `results/week3_table.csv`): **architecture** — EfficientNet-B0, ResNet-18 and ResNet-50 (4–24 M parameters) all reach val F1 ≈ 0.565 at 224 px, so the model is not the limit; **input representation** — 320 px improves F1 by +0.018 (12 runs), 384 px brings nothing more for 1.5× the time; **training strategy** — cosine is the only change that helps, augmentation and regularization hurt at a 5-epoch budget, 10–15 epochs overfit; **cost** — sections 7.1–7.4; **controlled comparison** — same splits, same budget, seeds 42–44.

6. Benchmarking methodology

- **Synchronization.** `GpuTimer` calls `torch.cuda.synchronize()` before and after every timed region. Without it an 8192^2 matmul "takes" 0.21 ms instead of 90.1 ms (`results/sync_demo.json`).
- **Warm-up.** `gpu_warmup()` keeps copies and matmuls in flight for 3 s before GPU benchmarks (a laptop GPU idles at PCIe Gen1); each end-to-end configuration has one discarded warm-up run (page cache, cuDNN autotuning, pretrained-weight download, cache build).
- **Timing boundaries.** *Setup* (config, loaders, cache build or GPU upload, precompute, model); *cold epoch* (epoch 0); *steady epoch* = mean of epochs 1–4, **train + validation**, transfers included; *total 5-epoch* = setup + cold + 4 × steady. One-time costs are always reported separately.
- **Repeats.** End-to-end: 3 timed repeats of the GPU run, 2 of the CPU baseline (285 s/epoch on one Colab core); ablation and test: 3 seeds each. We report mean ± standard deviation.
- **Profiling.** `benchmarks/profile_gpu.py` gives three views: (1) PyTorch Profiler (CPU + CUDA), 5 warm-up
  - 15 profiled steps, GPU busy time = **union** of kernel/copy intervals (not the sum, which over-counts overlapping streams), idle = 1 − busy / wall; (2) the same steps with a synchronization between data, forward, backward and optimizer, for attribution; (3) one full train epoch and one validation epoch. The profiler itself costs ~18 % (137.6 ms profiled vs 116.3 ms unprofiled step), so absolute times come from view (2)/(3) or from unprofiled runs, and the profiler is used for shares.
- **Quality.** Validation macro-F1 at the checkpointed epoch for every decision; test metrics once (7.3).

7. Results

7.1 Stage-level and end-to-end performance (Table 2)

Stage rows: 64 images, machine L, CPU vs GPU on the same machine (`results/timing.csv`). Epoch and total rows: machine

C, both configurations on the same machine (`results/week4/end_to_end.csv`).

Pipeline stage	CPU baseline (s)	Optimized (s)	Speedup	Optimized =
Image loading (64 img)	2.8078	0.0016	1,806.8×	decoded uint8 cache (CPU)
Preprocessing (64 img)	0.1582	0.0091	17.3×	batched GPU kernels
Filtering, 4 filters (64 img)	0.1944	0.0283	6.9×	GPU (torch); median 22.4× with the CuPy kernel at B = 128
Augmentation (64 img)	0.0306	0.0051	5.9×	GPU flip + rotation
Train step (64 img, reference CNN)	0.8837	0.0320	27.6×	GPU + AMP
Evaluation / inference (64 img)	0.2995	0.0132	22.8×	GPU + AMP
Training per epoch (steady, train + val)	285.41 ± 0.74	4.36 ± 0.02	**65.4×**	final configuration
One-time setup	0.3	1.7	0.2×	GPU upload + precompute (cache already built)
Total pipeline, 5-epoch training	1,427.5	23.5	**60.7×**	final configuration
Test-set inference (450 img, img/s)	19 (baseline model, GPU, CPU preprocessing)	686 (final model, GPU)	—	`results/week4/test_runs.csv`

The end-to-end rows compare *different models* (reference CNN at 224 px vs ResNet-18 at 320 px). To separate the GPU/pipeline effect from the model effect, the same machine C gives (`results/week3_table.csv`, steady epoch):

Step	Run	Epoch (s)	vs CPU	Val macro-F1
CPU reference (with cache)	`exp_cache_cpu`	200.28	1.0×	— (2-epoch timing run)
+ GPU, same model and data path (**device effect**)	`exp_cache_gpu`	7.83	25.6×	0.325
+ GPU pipeline, AMP, channels_last... (**pipeline effect**)	`optimized_cnn`	1.48	135.3×	0.305
+ ResNet-18, cosine, 320 px, batch 64 (**model effect**, final)	`optimized`	4.58	43.7×	0.572 ± 0.022

So the GPU and pipeline work make the *same* model 135× faster; the model redesign then spends 3.1× of that speed to raise quality from F1 0.31 to 0.57–0.59. The final design is still **65× faster per epoch than the baseline** on the same machine, and better.

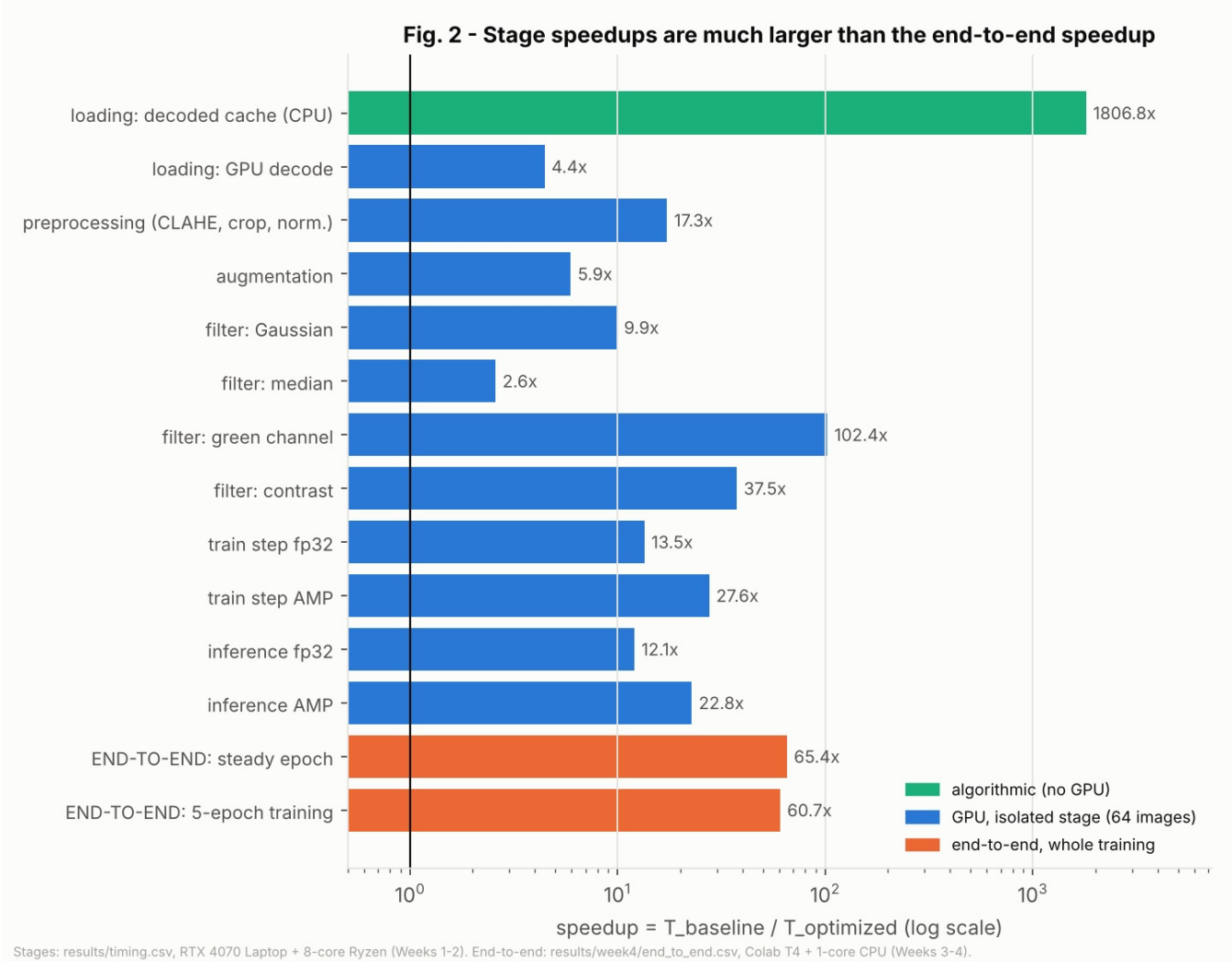


Figure 2 — Stage-level speedups (machine L) next to the end-to-end speedups (machine C).

Amdahl. The cache alone: p = 0.654, s = 1,807 → S = 1/((1 − p) + p/s) = **2.89×**, the bound itself; measured on machine L 1.53× for 5 epochs (the 21 s build and the remaining loader overhead are inside). All measured stage speedups together (loading 1,807×, preprocessing 17.3×, training step 27.6×, "other" 5.3 % unchanged) predict **15.5×**; the "other" 5.3 % alone caps it at 18.8×. The measured 135× for the same model is larger because (a) it was measured on machine C, whose single CPU core makes every CPU stage ~2.3× slower than on machine L (200 s vs 86 s per CPU epoch), and (b) the "other" Python/collate overhead disappears too once there is no DataLoader. This is the brief's point: a stage speedup says little about the application until the remaining fractions are known.

7.2 Where the time goes now: GPU profile of the final configuration (A7.2–A7.4, A7.7)

`python benchmarks/profile_gpu.py --config configs/optimized.yaml` (machine C, `results/week4/profile_optimized/`):

Quantity	Value
Training step (batch 64, 320 px, unprofiled)	**116.3 ms**, 550 img/s
GPU busy (union of kernels and copies) / idle	137.0 ms / step, **idle 0.4 %** (one gap > 50 µs every 15 steps)
Host→device copies per step	**none** (only 0.08 MiB device-to-device)
Kernels per step / mean duration	351 / 388 µs → not launch-bound
Synchronized phases	data 0.8 ms (0.6 %), forward 42.9 ms (35.6 %), backward 74.1 ms (61.6 %), optimizer 2.6 ms (2.1 %)
Epoch	train 3.76 s (32 steps) + validation 0.31 s (8 %) = 4.07 s

GPU kernel category	ms / step	Share
Convolutions (cuDNN implicit-GEMM, fp16 Tensor Cores, NHWC)	67.8	49.9 %
Batch norm (cuDNN NHWC fwd/bwd)	25.3	18.6 %
Elementwise (ReLU backward, residual add, clamp)	23.1	17.0 %
Max-pool (one layer: backward 10.2 ms, forward 6.7 ms)	17.0	12.5 %
Fused AdamW + AMP scaler	2.4	1.7 %
Data (randperm, `index_select` gather of the	0.4	0.3 %

resident batch)		

Reading the trace (A7.4). Between two training steps the GPU does not wait: the CPU issues kernels faster than the GPU executes them (21.7 ms of CPU time in `forward` for 42.9 ms of GPU forward work), so the queue never runs dry, and the only data work is a 0.4 ms gather from GPU memory. The bottleneck **moved** from the input path to the model's own kernels. Half of the GPU time is convolution, already on Tensor Cores in fp16 and NHWC; the other half is memory-bound layers (batch norm, elementwise, pooling) whose cost scales with activation size (320² inputs). The single max-pool layer costing 12.5 % is a concrete lead: its NHWC backward kernel is slow on the T4 (section 10).

Comparison with the DataLoader path (`--config configs/exp_resident_off.yaml`, `results/week4/profile_exp_resident_off/`): step 140.7 ms (+21 %); each step copies 21.3 MiB of uint8 from pinned memory (1.85 ms) and spends **23.7 ms** in GPU preprocessing (CLAHE at 320 px); the GPU is still idle only 0.4 %, so the single-core loader keeps up — what the resident + precompute design removes is GPU *work* (preprocessing every epoch), not GPU *waiting*. The ablation confirms it (next section).

7.3 Classification quality on the test set (Table 3)

The configuration was frozen after the profile (GPU idle < 10 %, no input-path bottleneck left) and the ablation. `scripts/week4_test_eval.py` then trained each model with seeds 42, 43, 44, checked on **validation** that evaluation reproduces the checkpoint's accuracy (gap 0.0000 for all six), and evaluated each checkpoint **once** on the 450 test images. A lock file records the single use (`results/week4/TEST_SET_USED.json`, 2026-10-29 02:47, configs' MD5). All seeds are reported, none chosen.

Metric (test, mean ± std over 3 seeds)	Baseline model	Final model
Accuracy	0.357 ± 0.014	**0.598 ± 0.020**
Macro-precision	0.358 ± 0.123	**0.604 ± 0.014**
Macro-recall	0.363 ± 0.014	**0.591 ± 0.023**
Macro-F1	0.313 ± 0.010	**0.590 ± 0.021**
Recall grade 0 / 1 / 2 / 3 / 4	0.60 / 0.43 / **0.01** / 0.30 / 0.47	0.46 / 0.69 / **0.42** / 0.55 / 0.84
Validation macro-F1 of the same checkpoints	0.327	0.562
Confusion matrix	Fig. 6 (`results/week4/confusion_baseline.json`)	Fig. 6 (`confusion_final.json`)

Macro-precision/recall/F1 are the unweighted means over the five grades of the one-vs-rest precision, recall and F1 (scikit-learn, `average="macro"`); with balanced classes accuracy is informative, but macro-F1 exposes the class the baseline ignores: **grade 2 is almost never predicted by the baseline (recall 0.01)**, as already seen on validation in Week 1. The final model predicts all grades.

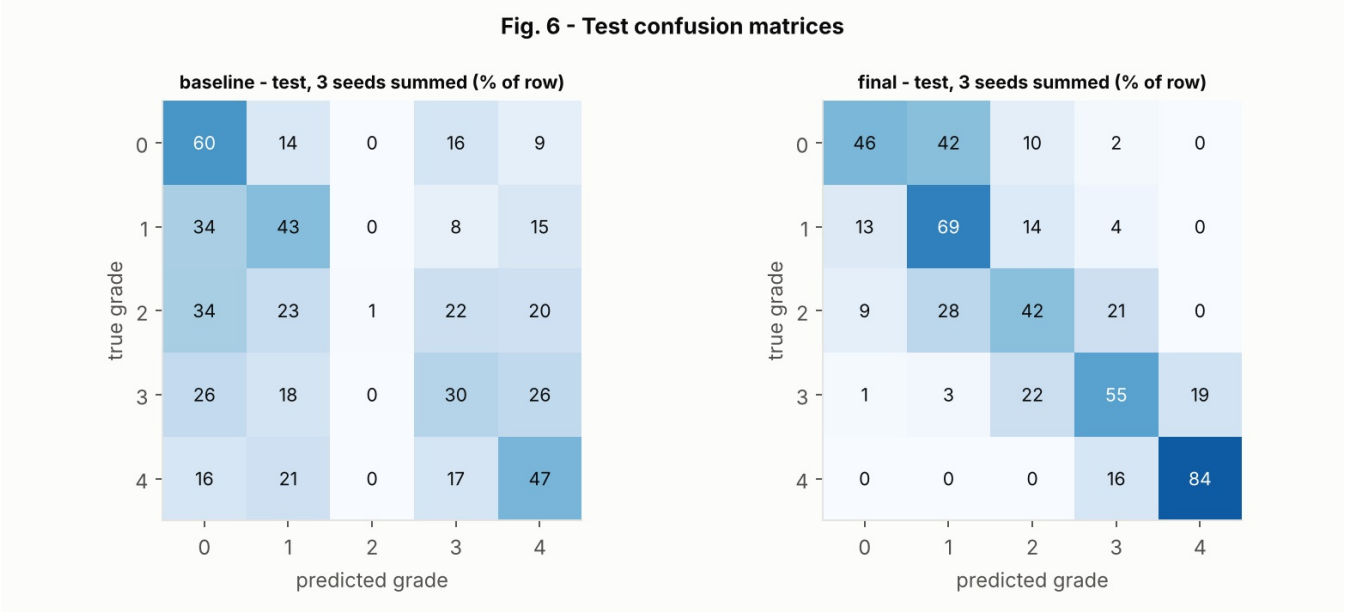


Figure 6 — Test confusion matrices, 3 seeds summed, % of each true grade.

The final model's errors are almost all **between adjacent grades** (0↔1, 1↔2, 2↔3, 3↔4): no grade-4 image is predicted as 0, 1 or 2, while the baseline sends 16 % of grade-4 images to grade 0. The weakest point is grade 0 (42 % predicted as grade 1). The test F1 (0.590) is slightly *above* the validation F1 of the same checkpoints (0.562) and within the seed spread: no sign that the ~20 configurations compared on validation in Week 3 over-fitted the validation set.

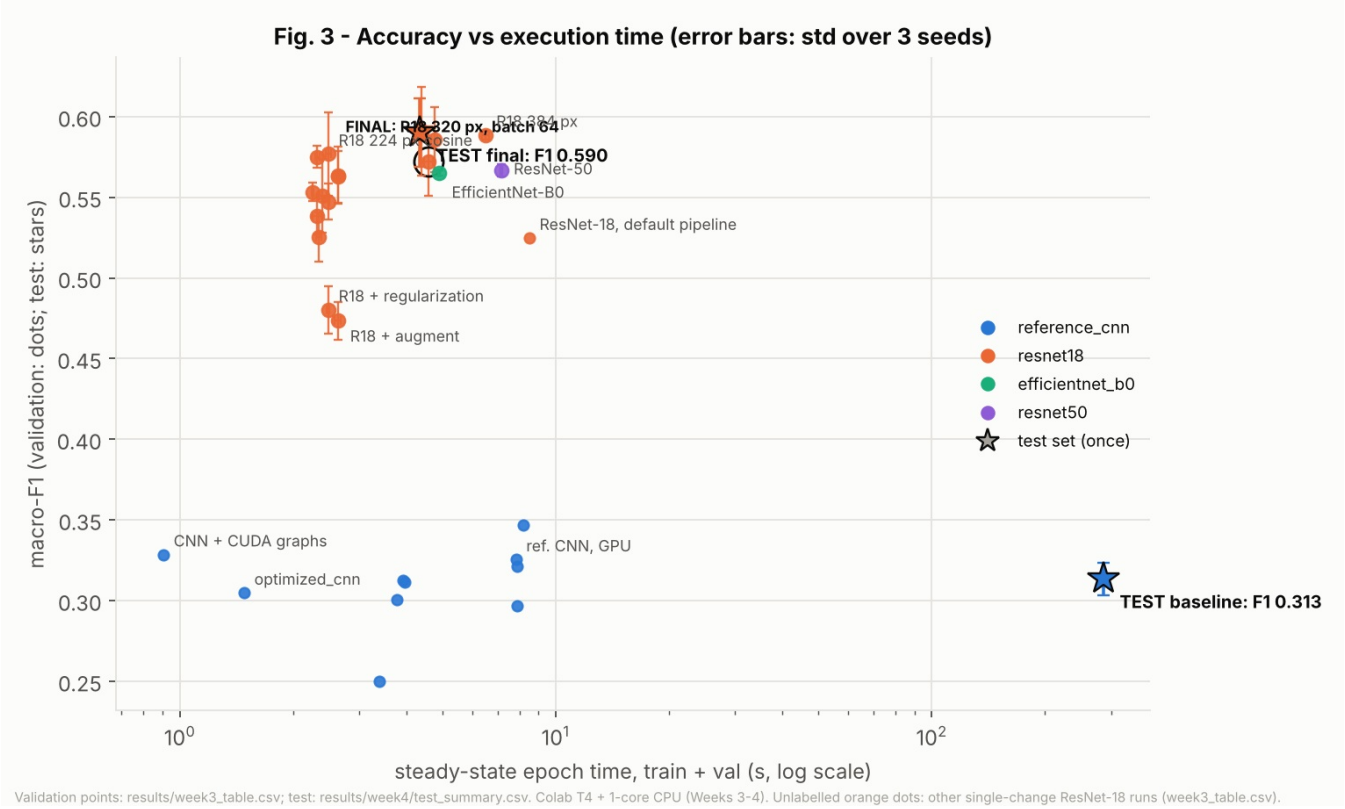


Figure 3 — Validation macro-F1 vs steady epoch time for every Week 3 configuration (dots), and the test F1 of the baseline and final model (stars).

7.4 Ablation (A7.6, Figure 4)

`scripts/week4_ablation.py`: from `configs/optimized.yaml`, **one optimization off at a time** (plus one clearly defined combination), 3 seeds, same GPU session, the full configuration re-run as reference.

Variant	Switched off	Steady epoch (s)	Slowdown	Peak GPU mem (MB)	Val macro-F1
`full`	nothing	4.35 ± 0.03	1.00×	3,357	0.576 ± 0.015
`amp_off`	mixed precision (fp32)	19.73 ± 0.03	**4.54×**	7,386	0.554 ± 0.054
`channels_last_off`	NHWC layout	5.70 ± 0.00	**1.31×**	3,354	0.578 ± 0.017
`resident_off`	GPU-resident data (+ precompute)	5.33 ± 0.04	**1.23×**	1,880	0.566 ± 0.020
`precompute_off`	preprocessing computed once	5.10 ± 0.01	**1.17×**	2,648	0.576 ± 0.015
`fused_adamw_off`	fused optimizer	4.40 ± 0.00	1.01×	3,334	0.553 ± 0.025
`cudnn_benchmark_off`	cuDNN autotuning	4.34 ± 0.00	1.00×	3,334	0.576 ± 0.015
`input_pipeline_off`	resident **and** GPU preprocessing (CPU CLAHE in workers)	16.09 ± 0.07	**3.70×**	1,877	0.566 ± 0.020

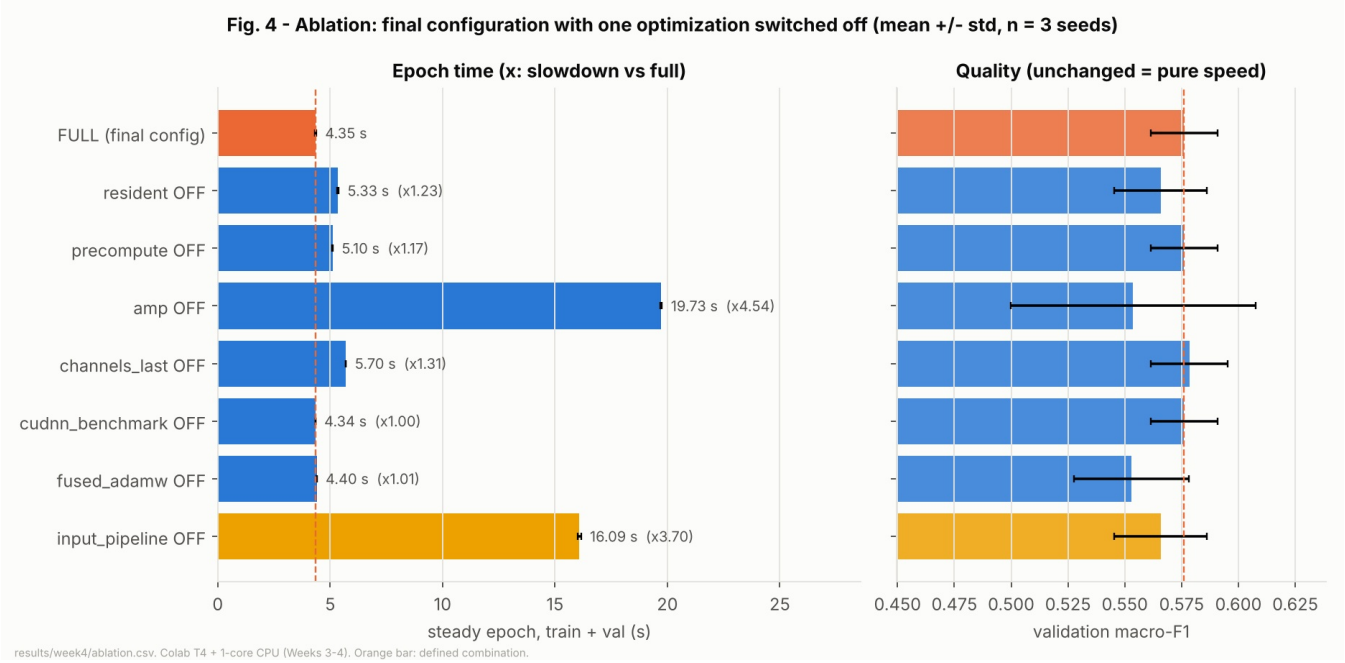


Figure 4 — Ablation from the final configuration (mean ± std over 3 seeds).

- **AMP is the largest single contributor (4.54×) and halves memory (7.4 → 3.4 GB).** The profile shows why: the step is compute-bound, and fp16 is the only way to use the T4's Tensor Cores (65 TFLOPS fp16 vs 8 TFLOPS fp32). In Week 3, while the run was input-bound, AMP alone gave nothing (8.20 vs 7.83 s): the same optimization is worth 0× or 4.5× depending on where the bottleneck is.
- `channels_last` **1.31×**: without it cuDNN inserts NCHW↔NHWC conversions around every Tensor-Core convolution.
- **Resident data 1.23× and precompute 1.17×**: they remove per-epoch GPU preprocessing (23.7 ms/step in the profile) and the copies; with CPU preprocessing instead (`input_pipeline_off`, 3.70×) the single Colab core becomes the bottleneck again. Cost: +1.5 GB GPU memory, acceptable on 16 GB.
- **Fused AdamW (1.01×) and cuDNN benchmark (1.00×) give nothing measurable.** The optimizer is 2.1 % of a step (Amdahl: even an infinitely fast optimizer gives ≤ 1.02×). For cuDNN, the default heuristics already choose the same kernels for these standard ResNet shapes; autotuning only adds cold-epoch time. Both are kept (harmless) but we do not claim a gain.
- **Quality is unchanged by speed optimizations, and some are bit-identical.** `full`, `precompute_off` and `cudnn_benchmark_off` give *exactly* the same F1 per seed: precomputing in fp16 changes nothing numerically. `resident_off` and `input_pipeline_off` are identical to each other — GPU and CPU preprocessing are bit-exact (Week 2) — and differ from `full` only through the batch order (DataLoader shuffle vs GPU `randperm`). `amp_off` and `fused_adamw_off` change floating-point arithmetic, hence different, not worse, results within the seed spread (seed 42 of `amp_off`: 0.49).

7.5 Resource utilization (Figure 5)

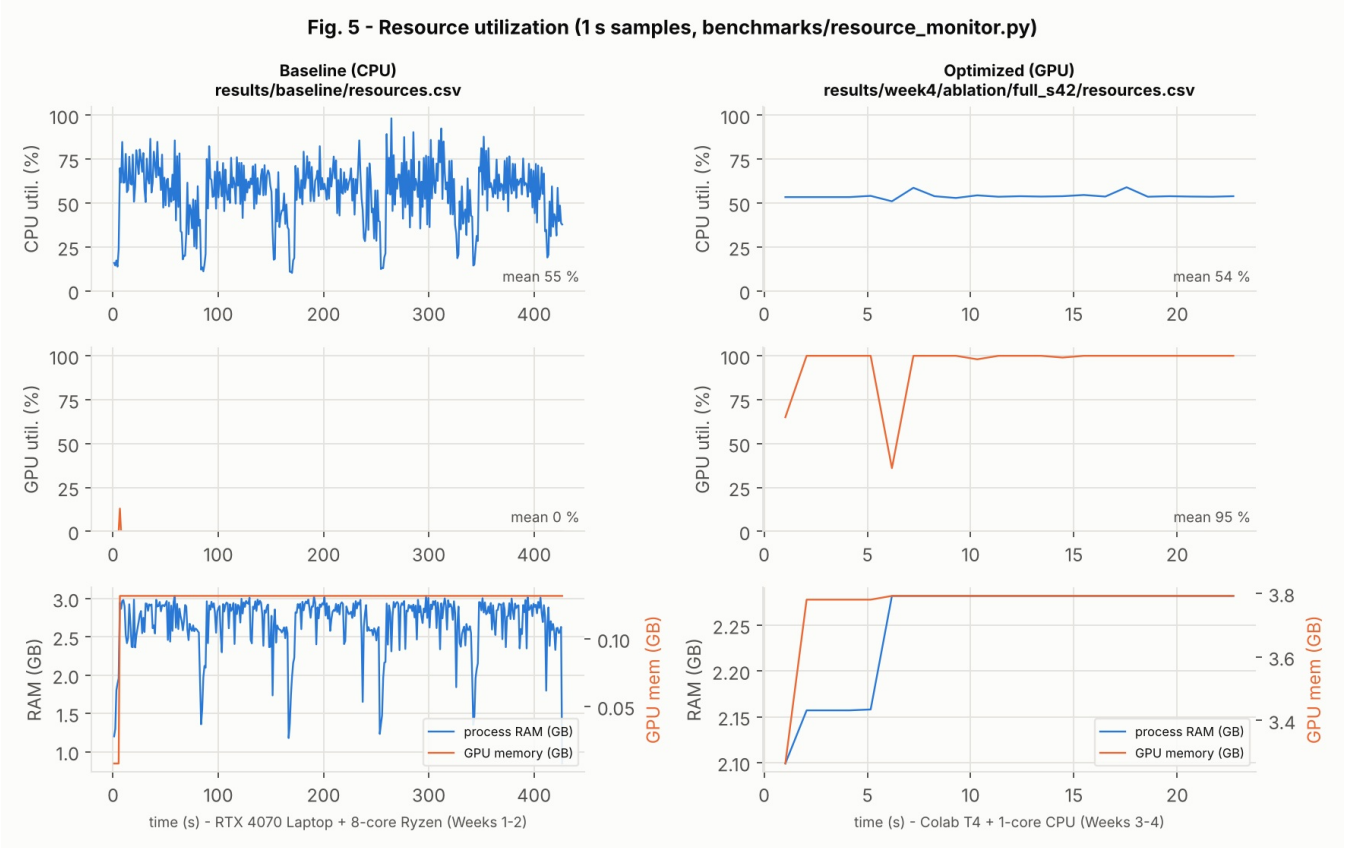


Figure 5 — CPU, GPU and memory over time: baseline (machine L) and final run (machine C).

The baseline uses the CPU at ~55 % on average with large dips (workers waiting on decode) and the GPU not at all. The optimized run keeps the **GPU at 95 % utilization** on average (nvidia-smi samples of `full_s42`; the dips are the validation pass and checkpoint saves), up to 3.9 GB of GPU memory (driver view) and 2.3 GB of host RAM; the single CPU core only launches kernels.

8. Discussion and critical analysis

Why did some stages benefit more than others? Per-pixel, embarrassingly parallel work (green channel 102×, contrast stretch 37.5×, normalization) gains most; neighbourhood operations with data reuse gain less unless written for the GPU (median: 2.6× with torch `unfold`, 22.4× with a custom CuPy kernel). Loading gained most of all, and **not from the GPU**: decoding was redundant, so the best "kernel" was to not run it again (1,807×). GPU JPEG decoding only reached 4.4×.

Did overheads reduce the gains? Yes, in two places measured in Weeks 2–3: kernel-launch overhead made the small reference CNN launch-bound (CUDA graphs gave 1.64× on it, nothing on ResNet-18, whose 351 kernels/step average 388 µs); and transfers, until data was made resident: 21.3 MiB per step is only 1.85 ms on PCIe, but the per-step GPU preprocessing behind it was 23.7 ms.

Did the bottleneck shift? Five times: JPEG decode (CPU) → CPU CNN (after the cache) → CPU preprocessing in the single-core loader (after the move to GPU) → GPU preprocessing (after GPU preprocessing, Week 3) → **GPU convolution / batch norm / pooling compute** (now, idle 0.4 %).

Were isolated gains reflected end-to-end? Only partially, as Amdahl predicts (sec. 7.1): 1,807× on loading became 1.53× end-to-end on machine L. Stage speedups of 6–1,800× combine into 65× per epoch.

Algorithmic vs hardware gains. On the same model and machine: device effect 25.6×; data-path redesign (GPU preprocessing, residency, precompute) 2.3×; precision and layout (AMP, channels_last) 2.3×; the cache is algorithmic and helps the CPU baseline as much as the GPU. The model change costs 3.1× in speed and buys +0.28 test macro-F1.

Classification quality. The gain in macro-F1 comes from the **model and input redesign** (pretrained ResNet-18 +0.20 val F1 at 224 px, cosine schedule +0.07, 320 px +0.018), not from GPU execution: the ablation shows speed optimizations leave F1 unchanged within noise. Preprocessing changes preserved retinal structures: GPU CLAHE is bit-exact with OpenCV, and the higher resolution helped precisely because lesions (microaneurysms, exudates) are small. The final evaluation used data never used for selection.

Engineering trade-offs. More GPU memory (3.4 GB, from 0.9 GB for the CNN; 7.4 GB without AMP); a 0.45–0.92 GB disk cache; custom GPU CLAHE code (more complex, CUDA-only; the CPU path is kept and selectable). Portability: the resident-data design needs the dataset to fit in GPU memory (it would not at 300,000 images, ~45 GB uint8: then cache + on-the-fly GPU preprocessing + prefetching is the right design). On a GPU with more fp32 throughput relative to fp16 (or without

Tensor Cores), AMP would gain less; on a machine with many CPU cores, the CPU baseline would be ~2.3× faster and our speedup smaller (machine L: 86 s vs 285 s per baseline epoch). The added complexity is justified by the measured gains of AMP, channels_last, residency and precompute; it is *not* justified for fused AdamW and cuDNN autotuning, which we keep only because they cost nothing.

9. Limits and threats to validity

- **Dataset.** 2,999 images (2,099 for training) from one source; balanced classes do not reflect clinical prevalence; the test set has 450 images (~90 per grade), so one image is 0.2 % of accuracy and the seed std (0.02) is of the same order as differences we discuss. Results say nothing about clinical safety or other cameras/hospitals; this is an educational benchmark, not a diagnostic device.
- **Two machines.** Stage tables (machine L) and end-to-end tables (machine C) are not comparable to each other; every speedup quoted compares runs on one machine. Machine C's single CPU core makes the CPU baseline weak: the same baseline epoch takes 86 s on machine L's 8-core CPU vs 285 s here, so against a multi-core CPU the ratio would be roughly 3.3× smaller (an estimate, not a measurement).
- **Hardware state.** Colab assigns a different VM per session; the T4 SM clock read 825 MHz in P0 after warm-up (power/thermal capping is common on Colab). The ablation re-ran its reference in the same session for this reason; Week 3 and Week 4 numbers for the same config differ by 5 % (4.58 vs 4.35 s).
- **Measurement.** The profiler adds ~18 % to a step; shares are robust, absolute profiled times are not. CPU baseline end-to-end: 2 repeats only (285 s/epoch); total 5-epoch times are extrapolated from 2-epoch runs (setup + cold + 4 × steady).
- **Numerics and seeds.** AMP and fused AdamW change floating-point results; GPU training with `cudnn_benchmark` can be non-deterministic (Week 3 saw ±0.02 F1 between identical configs). Three seeds establish directions, not statistical significance for differences below ~0.03 F1.
- **Selection.** About 20 configurations were compared on the same 450 validation images; the test result (above validation) does not suggest over-fitting, but n = 450 limits what it can show.
- **Baseline model trained on the GPU for its test metrics** (same config, fp32, CPU preprocessing): only the device changes; Week 3 showed the device does not change what the model learns beyond seed noise.

10. Conclusion and future work

Measuring first changed every decision: the obvious target (move the CNN to the GPU) was not the first bottleneck; JPEG decoding was, and a CPU cache removed it. Each later optimization paid only once the bottleneck had moved to it (AMP: 0× while input-bound, 4.5× once compute-bound). The final pipeline is **60.7× faster end-to-end** than the baseline on the same machine, **doubles test macro-F1 (0.31 → 0.59)**, keeps the GPU busy 99.6 % of the time, and every claim is backed by a file in `results/`.

Future work, from the remaining profile: (1) replace the stem max-pool (12.5 % of GPU time; its NHWC backward is slow on the T4) by a strided convolution or test an NCHW pool inside the NHWC model; (2) fuse batch-norm + ReLU + add (35 % of GPU time is memory-bound) with `torch.compile` at this resolution, measuring the cold-epoch cost; (3) inference-only optimization (TensorRT / fp16 export); (4) an ordinal loss, since errors are between adjacent grades; (5) the same protocol on another GPU (the team's RTX 4070) to separate hardware-specific from general conclusions.

11. Reproducibility

Code: `README.md` lists every command. Week 4 runs on Colab T4 with `notebooks/week4_colab.ipynb`:

```
python scripts/download_data.py                      # FDR26 from Zenodo, MD5-checked
python -m pytest tests -q                            # CPU/GPU equivalence, prefetcher
python benchmarks/profile_gpu.py --config configs/optimized.yaml        # sec. 7.2
python benchmarks/profile_gpu.py --config configs/exp_resident_off.yaml --no-epochs
python scripts/week4_ablation.py --seeds 42,43,44                       # sec. 7.4
python scripts/week4_test_eval.py                                       # sec. 7.3 (once)
python benchmarks/benchmark_end_to_end.py --only optimized_gpu --epochs 2 --repeats 3
python benchmarks/benchmark_end_to_end.py --only baseline_cpu --epochs 2 --cpu-repeats 2
python scripts/make_figures.py                                          # figures + results/tables.md
```

Appendix — Checkpoint answers (A7.9) and defense questions (8E)

Stage speedup 30× but end-to-end 3×? If the stage is a fraction p of the run, S = 1/((1 − p) + p/30); S = 3 gives p ≈ 0.69: the other 31 % is untouched and bounds the result at 1/0.31 ≈ 3.2×. Our own case: loading 1,807× with p = 0.654 → 2.89× bound, 1.53× measured including the cache build.

Which optimization helped least relative to expectation? cuDNN autotuning and fused AdamW (1.00×, 1.01×). The profiler explains it: the optimizer is 2.1 % of a step, and cuDNN's heuristics already pick the implicit-GEMM fp16 NHWC kernels autotuning would pick. Earlier, the prefetcher (no gain: the uint8 copy it hides was < 2 % of a step).

Defense, six questions. (1) Principal bottleneck: JPEG decode + resize, 65.4 % of the baseline epoch (cProfile, `profile_stages.json`). (2) Greatest end-to-end improvement: on the CPU, the cache; on the GPU, mixed precision (4.54× in the ablation), after the data path was fixed. (3) Less than expected: prefetcher, fused AdamW, cuDNN autotuning, and AMP

while input-bound. (4) Quality: test macro-F1 0.313 → 0.590, from the model/input redesign, not from the speed optimizations (ablation). (5) Remaining bottleneck: GPU compute — convolutions 50 %, batch norm 19 %, elementwise 17 %, max-pool 12.5 %; GPU idle 0.4 %. (6) Reproducible elsewhere: correctness (bit-exact preprocessing, test metrics within seed noise), the bottleneck sequence and the Amdahl reasoning; hardware-dependent: absolute times, the 65× ratio (1-core Colab CPU), the size of the AMP gain (T4 Tensor Cores), memory headroom for resident data.